

Ryan Joseph Patrick Casalino

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Summary

Senior Software & Systems Engineer specializing in high-availability public infrastructure and containerized Go microservices. Proven track record of driving zero-downtime blue/green deployments, managing AWS infrastructure at scale, and architecting resilient systems from low-level declarative OS configurations up to high-concurrency API gateways.

Technical Core

[Go](#) [C/C++](#) [Python](#) [Nix / NixOS](#) [TypeScript](#) [PostgreSQL](#) [AWS \(ECS, S3, Aurora\)](#) [OpenTofu](#) [Docker](#) [gRPC](#) [Kong API](#)

Experience

Ad Hoc LLC

2022 – Present

Seattle, WA

Senior Software Engineer

- Architect high-throughput, containerized Go microservices powering Medicare.gov Plan Compare, serving **70M+ eligible beneficiaries** nationwide during high-concurrency Annual Open Enrollment periods.
- Maintained an uninterrupted 5-year production track record (2012–2026) using grpc-gateway to marshal millions of daily REST/JSON web requests into low-latency gRPC backend microservices with zero framework failures.
- Optimize API latency and database performance across Aurora PostgreSQL and Redis caching layers to sustain peak traffic loads behind Kong and Akamai API gateways.
- Execute zero-downtime blue/green deployments utilizing LaunchDarkly feature flags while maintaining 24/7 operational readiness across New Relic, Splunk, SonarQube, and JFrog Xray pipelines.

Sinch

2020 – 2022

Bothell, WA

Software Engineer

- Engineered an automated, headless video rendering pipeline using Ruby and Bash to dynamically orchestrate AWS EC2 Windows instances executing Adobe After Effects CLI tasks.
- Scaled personalized media generation workflows to render custom video/graphic assets on-demand for high-volume SMS and MMS customer engagement campaigns.
- Documented and refactored legacy Content Management Systems, mapping critical dependencies to streamline team workflows and reduce engineering onboarding time.

Selected Projects & Independent Engineering

2017 – 2020

Seattle, WA

Systems & Software Engineer

- **Declarative Infrastructure Provisioning:** Authored OpenTofu scripts and layered Docker images to deploy web applications and ACM SSL certificates across AWS ECS infrastructure.

- **gRPC Data Services:** Developed a Go-based gRPC service to asynchronously fetch, process, and persist bulk metadata from remote REST APIs into a normalized PostgreSQL schema.
- **Embedded Systems & Wayland:** Cross-compiled custom NixOS images for single-board computers (Broadcom/Allwinner SoCs), configuring Wayland compositors (Sway) for hardware-accelerated kiosk environments.

Education

City University of Seattle

Expected 2026

Seattle, WA

Master of Science in Computer Science

School of the Art Institute of Chicago

2011

Chicago, IL

Bachelor of Fine Arts